Pablo Luesia-Lahoz

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Lecturer and PhD Candidate at Graphics and Imaging Lab, Universidad de Zaragoza (Spain)

Education

Nov 2021 - Present

Universidad de Zaragoza, *PhD candidate*
PhD candidate supervised by Adolfo Muñoz and Diego Gutierrez. On computer graphics, computational imaging, Non-Line-of-Sight imaging, and transient imaging. 175 hours of experience in teaching undergraduate and master's students with a highly positive assessment.

Sep 2019 - Dec 2020

Universidad de Zaragoza, *Master's Degree*
Master's Degree in Computer Engineering. Average grade: 8.25 out of 10

Sep 2014 - Sep 2018

Universidad de Zaragoza, *Bachelor's Degree*
Bachelor's Degree in Computer Engineering with a major in Computing. Average grade: 7.64 out of 10

Research Experience

Apr 2024 - Jul 2024

University of Wisconsin-Madison, *International Internship*
International internship under the supervision of prof. Andreas Velten at the University of Wisconsin-Madison, and in collaboration with the Computational Optics Group. The research focuses on pushing the boundaries of NLOS imaging for looking around corners.

Nov 2021 - Present

Graphics & Imaging Lab* (Universidad de Zaragoza - I3A), *PhD candidate
PhD candidate under the supervision of Prof. Adolfo Muñoz and Prof. Diego Gutierrez. The research focuses on transient illumination, looking around corners, and virtual wave optics. Funded by *Ministerio de Ciencia e Innovación (Gobierno de España)* with the competitive grant FPI 2020. *Taught 175 hours of teaching to undergraduate students with a highly positive assessment.*

Jan 2021 - Jun 2021

Universidad de Zaragoza, *Research project collaboration*
E. Coli bacteria simulation for the production of citramalate in the *Instituto Universitario de Investigación en Ingeniería de Aragón (I3A)*. Detailed achievements:

- Learned how to use a cluster for heavy simulations.
- Acquired experience with the design and implementation of virtual cell models.

Sep 2019 - Dec 2020

Universidad de Zaragoza, *Research Project Intership*
Design and implementation of a system for the automatic analysis of the drone sperm cells of drones (honey bees). Funded by a competitive scholarship granted by the *Gobierno de Aragon*, in the *Instituto Universitario de Investigación en Ingeniería de Aragón (I3A)* for the Master Thesis Project.

Publications

Journals

SJR Q1

Non-Line-of-Sight in the Presence of Scattering Media
Pablo Luesia, Miguel Crespo, Adrian Jarabo, and Albert Redo-Sanchez. Optics Letters 47.15 (2022): 3796-3799.
DOI: `https://doi.org/10.1364/OL.463296`

SJR Q1

Cohesive Framework for Non-Line-of-Sight Imaging based on Dirac Notation
Albert Redo-Sanchez, Pablo Luesia-Lahoz, Diego Gutierrez, Adolfo Muñoz. Optics Express 32.6 (2024): 10505-10526.
DOI: `https://doi.org/10.1364/OE.518466`

SJR Q2

Stereo Non-Line-of-Sight Imaging
Pablo Luesia-Lahoz, Sergio Cartiel, Adolfo Muñoz. The Visual Computer (2026) 42:148
DOI: `https://doi.org/10.1007/s00371-025-04340-7`

Conference

Core B

Zone Plate Virtual Lenses for Memory-Constrained NLOS Imaging
Pablo Luesia-Lahoz, Diego Gutiérrez, and Adolfo Muñoz. ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Rhodes Island, Greece, 2023, pp. 1-5
DOI: `https://doi.org/10.1109/ICASSP49357.2023.10094929`

Core B

Zero-Phase Phasor Fields
Pablo Luesia-Lahoz, Talha Sultan, Forrest B. Peterson, Andreas Velten, Diego Gutierrez, Adolfo Muñoz. 2025 IEEE International Conference on Computational Photography (ICCP), Toronto, ON, Canada, 2025, pp. 1-9
DOI: `https://doi.org/10.1109/ICCP64821.2025.11143850`

Teaching and Supervision

Teaching

Experience in 175 hours at the University of Zaragoza (*Universidad de Zaragoza*):

2022 - 2025

Computer Graphics (158 hours)
Degree on Computer Sciences Engineering. Received an outstanding positive assessment from the students.

2024 - 2025

Fundamentals of Computing (7.5 hours)
Master's in Robotics, Graphics and Computer Vision in the University of Zaragoza *Universidad de Zaragoza*. Received an outstanding positive assessment from the students.

2022 - Present

Fundamentals of Computing (10 hours)
Degree in Industrial Technology Engineering

2025 - Present

Fundamentals of Computing

2025 - Present

Programming

2025 - Present

Current Students

Present

Ming Tao Ye
Internship "On the Rendering of Optical Transparency".

Former students

2024

Jorge Solán Morote
Bachelor thesis. "Simulation of ultra-fast photon capture hardware in light transportation." Graded with a 9 out of 10.

2023

Isaac Velasco
Internship "Geometry optimization using normal information in Non-Line-of-Sight imaging".

2023

Ayush Gupta
Internship "Evaluation of the Fourier components in a Transient Signal".

Divulgation activities

8-11 Aug 2022

Attendance to ACM SIGGRAPH 2022
Presentation of the poster "Non-line-of-sight Transient Rendering".

30 Sep 2022

European Researchers' Night
Divulgation experiments for a general public of all ages: "Create your own virtual character".

15 July 2023

***XII* Jornada de Jóvenes Investigadores/as**
Presentation of the poster "Zone Plate Virtual Lenses for Memory-Constrained NLOS Imaging".

29 Sep 2023

European Researchers' Night 2025
Divulgation experiments for a general public of all ages: "Create your own virtual character", "Learn about Virtual Reality", and "How is color composed?".

18-22 Mar 2024

XV Engineers Week
Divulgation activities for a high-school public to present the research of the Graphics & Imaging lab.

20 Mar 2024

XII Aragon Girls' Day
Divulgation activities in collaboration with the Graphics & Imaging lab to a high-school-aged public focused on making visible the researcher women.

21-23 July 2025

Presentation in ICCP conference

Attendance and presentation of our work "Zero-Phase Phasor Fields for Non-Line-of-Sight Imaging" in the 2025 IEEE International Conference on Computational Photography (ICCP) held in Toronto, ON, Canada.

26 Sep 2025

European Researchers' Night 2025

Divulgation experiments for a general public of all ages.

Volunteer Experience

2014 - present

Aragón (Spain), Free Time Monitor

Experience in camps with kids from 8 to 16 years old. Approved Free Time Monitor Title by the Instituto Aragones de la Juventud (IAJ), in Spain.

Other interest data

Programming languages

- Java
- C++
- Python
- Matlab
- Haskell
- ProLog

Languages

- **Spanish** Mother language
- **English** Advanced (CERT C1 certificate emitted by the British Council with the Aptis ESOL)
- **French** Basics

Mobility

- Erasmus stay in Sweden for 6 months (Jan 2019 - Jun 2019).
- Internship in Madison (Wisconsin) for 3 months (May 2024 - July 2024).

Interests

Professional

Research in the computational imaging field, transient luminescence and rendering, non-line-of-sight and virtual wave optics

Personal

Cooking, rock climbing, reading, guitar, drawing